V18 Germanium detector in γ-spectroscopy

Abstract

In the experiment, the energy calibration of the detector and a measurement of the full energy probability as a function of the energy should first be carried out. The collected data will then be used to evaluate spectra of unknown emitters with regard to energy and activity.

References

[1] G. Knoll *Radiation Detection and Measurement*, Wiley 2010

[2] G. Gilmore *Gamma-ray Spectrometry*, Wiley 2008

[3] H. Kolanoski, N. Wermes *Particle detectors*, Springer 2016

[4] G. Lutz, *Semiconductor Radiation Detectors* Springer 2007

[5] S.M. Sze, *Physics of semiconductor devices*, Wiley 2007

Preparation

After preparing, you should be able to answer the following questions.

- By what process do photons interact with matter? Which processes dominate at small or very large energies (relevant to nuclear physics)? Which material properties have an influence on the probability of interaction? How is this related to the extinction coefficient? How does this depend on energy?
- What is the prediction for the differential cross section as a function of energy for Compton scattering?
- What physical size of the semiconductor limits the line width of gamma lines? Compare direct and indirect semic onductors. Which experimental effects lead to further line propagation?
- How are the signals from a germanium detector amplified? What is the difference between charge and current amplifiers?
- How are energy measuring detectors calibrated?
- How can the full energy probability of germanium detectors be determined using a point source at a large distance?
- What characteristic structures can be observed in a monochromatic gamma spectrum?
- What is the photon energy with the greatest probability of emission for ^{152}Eu, ^{137}Cs and ^{133}Ba?

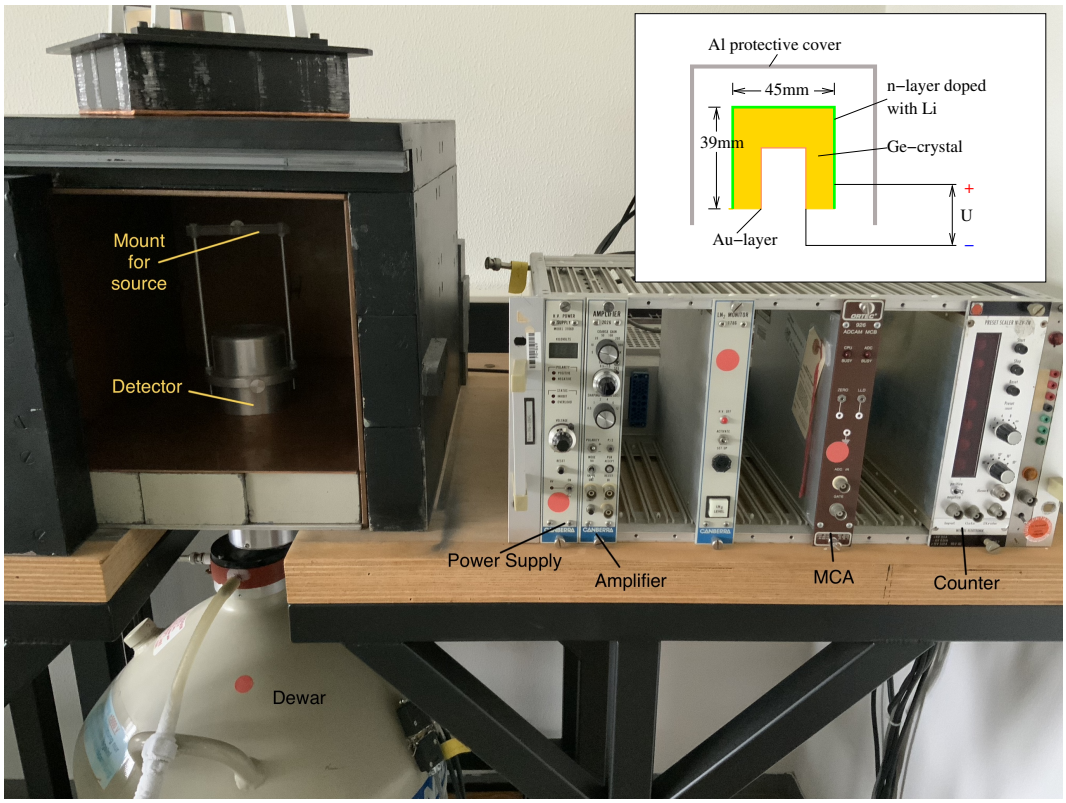


Figure 1: Experimental setup with sketch of the Ge-detector (inset)

Experimental setup and Alignment

The coaxial Ge detector used here has the shape of a cylinder. The surface ($d_{Li} = 0.5\ cm$) is n-doped by diffusion of Li atoms and is therefore highly conductive ($n \approx 10^{10} atoms/cm^3$). It is used to connect the positive pole of the reverse voltage. Inside the detector crystal there is a coaxial hole whose (inner) surface is vapor-coated with gold ($d_{Au} = 20\ \mu m$). This metal-semiconductor contact corresponds to a strong p-doping, so that an extensive depletion zone is formed because the acceptor density in the ge crystal is very small. The entire detector crystal is still surrounded by a protective bonnet made of Al sheet, so that the incident gamma must penetrate both the Al layer and the Li-doped surface in order to be detected. As a result, there is a lower detection limit for the gamma energy. For the detector used here, it is 40 keV to 50 keV. For this reason, only energies greater than 150 keV should be considered for determining the full-energy detection probability.

Measurements and analysis

Analyse the data from the following point of view:
Energy calibration and determination of the full-energy detection probability: Record the spectrum of a calibrated ^{152}Eu source to perform an energy calibration of the apparatus. Measure the full-energy detection probability of the detector. Both the position

of the lines and their content must be determined. To do this, compile the known gamma energies of the ^{152}Eu spectrum and the associated emission probabilities. Then perform a linear compensation calculation with the data for the energy calibration of the spectrometer. The ^{152}Eu source had an activity of (4130 ± 60) Bq on the day it was produced (01.10.2000). Calculate from this the activity of the source on the day of measurement. Calculate the present solid angle from the experimental setup. Take into account the distance between the detector and the aluminium protective bonnet, which is 1.5 cm. Furthermore, take the line contents of the individual lines from the spectrum and calculate the full energy detection probability of the lines of the ^{152}Eu spectrum.
Plot the full energy detection probability as a function of energy and fit a power function.
Determine the line content of the spectral line and calculate the activity of the measured source on the day of the measurement.

Investigate a monochromatic γ-spectrum: Determine from the spectrum the energy of the ^{137}Cs emitter, the half-width and the tenth-width of the full energy line, the position of the Compton edge and the backscatter line as well as the content of the full energy line and the Compton continuum.
The content of the Compton continuum is distored by the backscatter line and possibly by other influences in the low energy range. For a more precise determination, it is useful to adjust it by a function according to the differential cross section. A constant factor can be obtained from measurement results slightly below the Compton edge, because in this energy range the Compton curve is measured largely unbiased.
Compare the measured values for the Compton edge and the backscatter line with the values calculated from the energy. Check wether the measured half-value width of the photoline is consistent with the tenth-value width (What is the relationship between these two quantities for a Gaussian distribution?)
From the length of the detector crystal and the extinctions coefficient for photo and Compton effect, determine the absorption probability of the ^{137}Cs-radiation for the detector here. What conclusion should you get from the comparison? How does the photoline presumably come about?

Activity determination: Measure the spectrum of a ^{133}Ba source and determine the line contents of the individual spectral lines and calculate the activity of the measured source on the day of the measurement. Again, note the different emission probabilities.

Unknown source: Nuclide idetification and activity determination: Determine the position and content of the individual spectral lines. Identify which active isotopes occure using the given database. If these are obtained from the natural decay series, determine the members of the decay series that occure. Determine the activity of the individual isotopes.